



Manufacturing and Characterization of Polymer-Derived Ceramic Matrix Composites for High-Temperature Aerospace Applications

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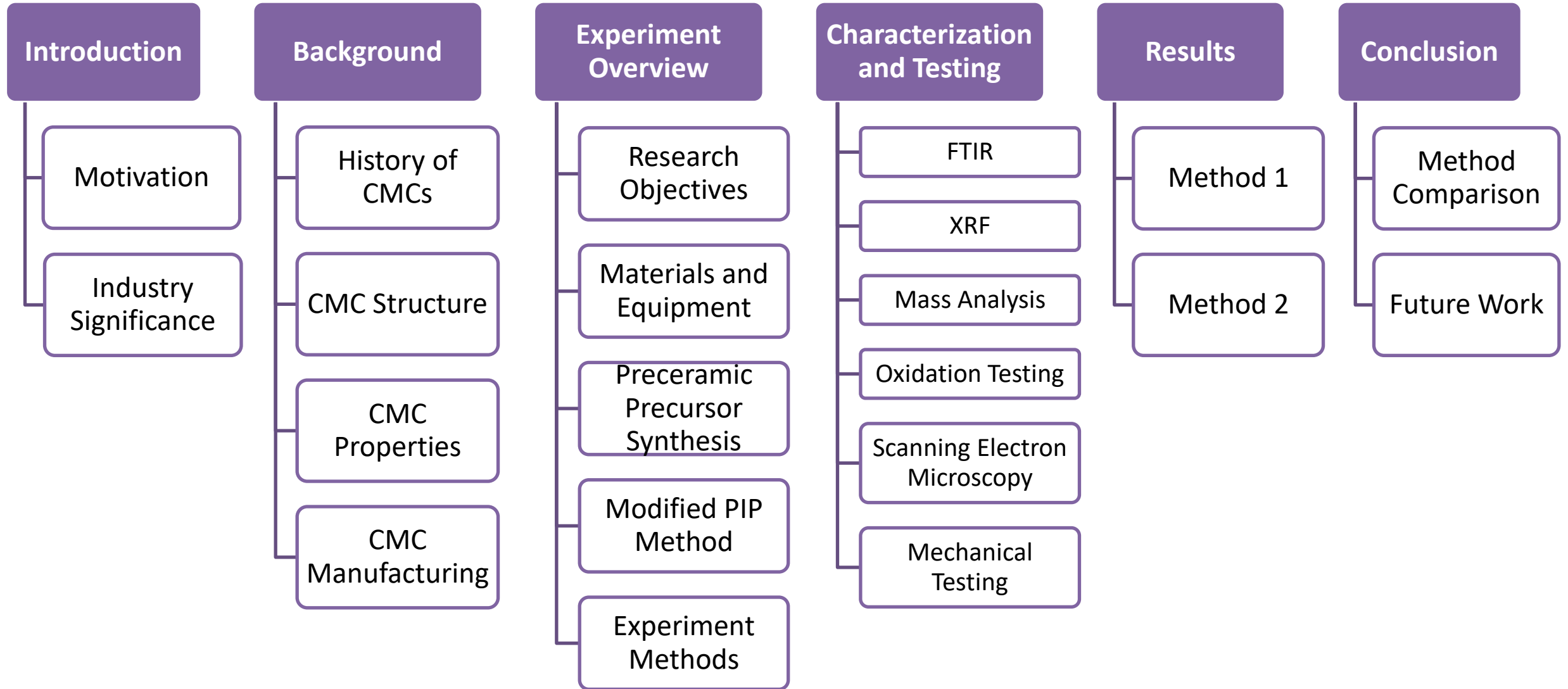
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Outline



Motivation

Problem: Temperature Limits Performance

Aerospace propulsion systems operate at $>5000^{\circ}\text{F}$

Higher temperatures $\rightarrow$ higher efficiency & specific impulse

Metals require cooling $\rightarrow$ added weight, complexity, and efficiency loss



OPPORTUNITY: Advanced Ceramic Materials

- Stable at extreme temperatures
- No active cooling required
- High oxidation and creep resistance
- Potential lighter, more efficient systems

LIMITATIONS: Implementation

- Brittle behavior
- Reliability concerns
- Difficult, costly manufacturing

GAP

- Materials that combine:
- High-temperature performance
 - Mechanical reliability
 - Scalable manufacturing

THIS WORK

- Explore ceramic-based systems
- Investigate processing methods
- Improve performance and manufacturability

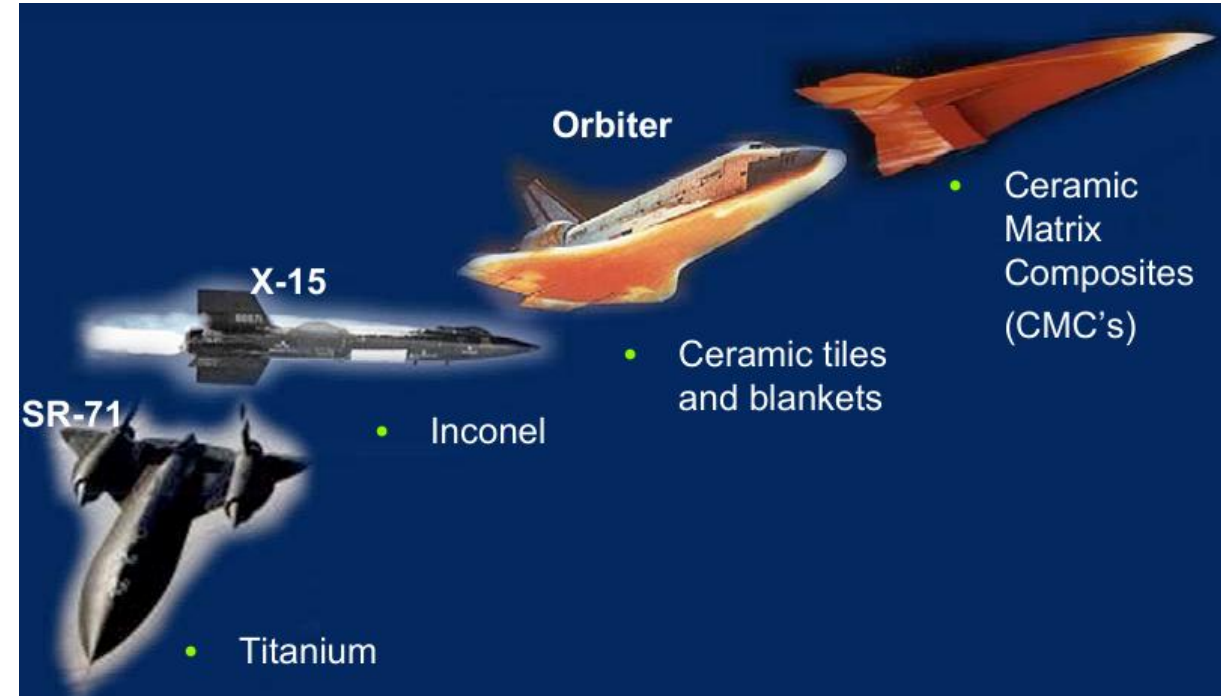
Industry Significance

Applications

- Jet turbines
- Thermal protection systems
- Hypersonic structures
- Rocket nozzles

Impact of this work

- Higher efficiency propulsion systems
- Longer-lasting high-temperature components
- Scalable manufacturing for aerospace applications

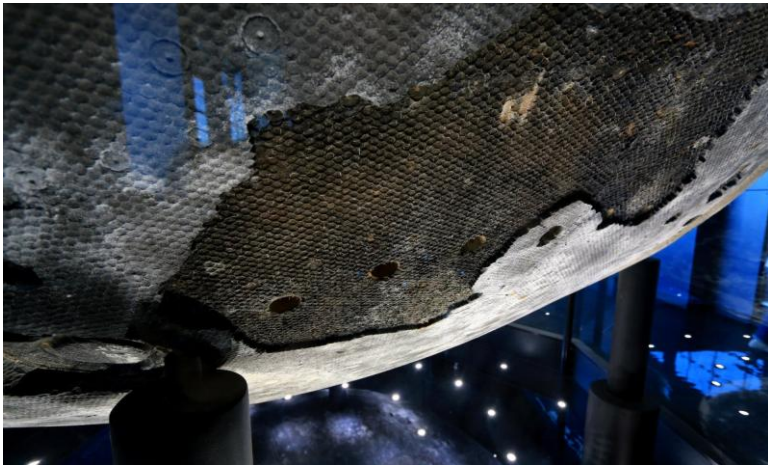


BACKGROUND

History of Ceramic Matrix Composites

Early Development (1950s-1970s)

- Monolithic ceramics used for high-temperature applications (brittle failure limitations)
- Early reinforcement attempts using ceramic fibers
- Initial aerospace interest



Advancement Phase (1980s-1990s)

- Development of fiber-reinforced ceramic systems (CF/SiC, SiC/SiC)
- Introduction of fiber-matrix interface engineering
- Emergence of CVI and PIP processing methods



Modern Applications (2000s-Present)

- Adoption in aerospace: turbine engines, hypersonics, reentry vehicles
- Improved oxidation-resistant systems
- Focus on scalable, cost-effective manufacturing



Traditional Ceramics vs. Ceramic Matrix Composites



Traditional (monolithic) ceramic



Ceramic matrix composite

Ceramic Matrix Composites Structure

Primary components

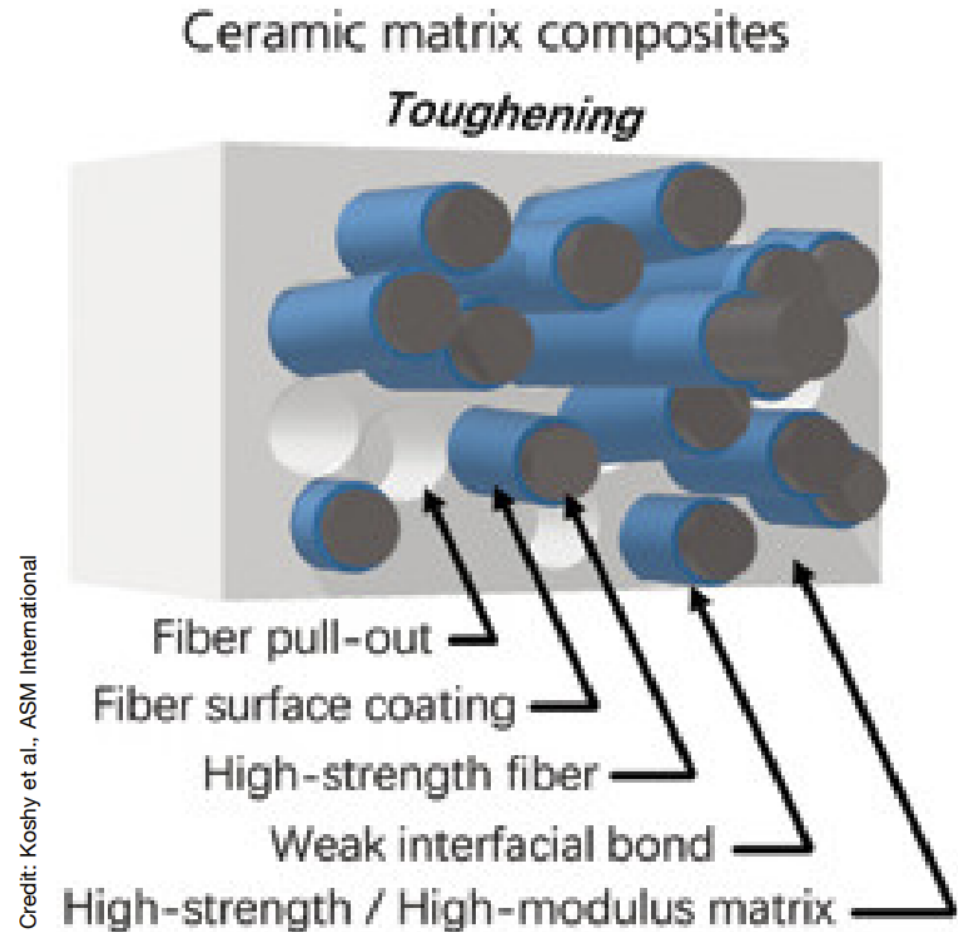
- Fiber reinforcement: Provides strength and toughness
- Ceramic matrix: Provides thermal stability and environmental resistance
- Fiber-matrix interface: Enables crack deflection and energy dissipation

Failure mechanisms

- “Inverse” composite: matrix cracks first
- Crack deflection at interface
- Fiber bridging and pullout → non-brittle behavior

Architecture

- 1D (unidirectional), 2D woven, 3D braided preforms
- Tradeoff: manufacturability vs. mechanical performance



Ceramic Matrix Composites Properties

Thermal Properties

- Stable at 1200-1600°C
- Low thermal expansion
- Strength retained at extreme temperatures

Mechanical Properties

- High strength-to-weight ratio
- Non-brittle, localized failure behavior
- Moderate strength and stiffness relative to metals

Environmental Resistance

- Excellent oxidation, corrosion, and creep resistance at high temperatures
- Suitable for harsh environments for aerospace and energy industries

Key Limitations

Property variability
(fiber/matrix/interface dependent)

Higher cost than
traditional materials

Processing-dependent
porosity and defects

Ceramic Matrix Composites Manufacturing

Chemical Vapor Infiltration

Fiber preform

Gas infiltration

Chemical reaction

Matrix Deposition

Benefits:

- High purity, uniform matrix
- Excellent high temperature performance
- Minimal residual stresses

Challenges:

- Very slow process
- High cost and equipment complexity
- Limited scalability

Highest performance vs. lowest manufacturing efficiency

Melt Infiltration

Porous preform

Molten silicon infiltration

Reaction (SiC formation)

Densification

Benefits:

- Fast processing time
- High density, low porosity
- Good mechanical strength

Challenges:

- Limited to specific material systems
- Residual phases (free Si)
- Less microstructural control

Fast and dense vs. limited material flexibility

Polymer Infiltration and Pyrolysis

Liquid precursor, fiber preform

Crosslinking

Infiltration

Pyrolysis

Benefits:

- Low-cost, scalable processing
- Compatible with complex geometries
- Tailorable chemistry

Challenges:

- Shrinkage during pyrolysis, porosity
- Lower density compared to other methods

Scalability and flexibility vs. density and porosity

EXPERIMENT OVERVIEW

Research Objectives

Primary Goal: Develop and evaluate a modified PIP process for CF-reinforced SiCN and SiBCN mini-composites

Hypothesis: Repeating infiltration + crosslinking before single pyrolysis 1.) improves coating uniformity, 2.) reduces processing time and 3.) maintains comparable material performance

Comparison

- Method 1: Traditional multi-cycle PIP
- Method 2: Modified multi-cycle (single pyrolysis)

Materials Selection

- SiCN: baseline polymer-derived ceramic
- SiBCN: boron-modified system for improved oxidation behavior

Key Evaluation Metrics

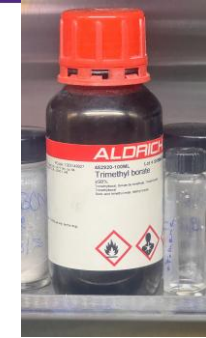
- Ceramic yield
- Oxidation resistance
- Microstructure (SEM)
- Mechanical properties

Materials and Equipment

- Reinforcement: Carbon fiber cloth
 - Fibre Glast 6K, 2x2 twill weave, 0.017in thickness, 10.9 oz/yd² areal weight
- Preceramic polymer
 - Poly(ureamethylvinyl)silazane
 - Ceraset PSZ20, Clariant Corporation
 - Isopropanol
 - ≥99.9%, MW 60.096 g/mol, Fisher Scientific
 - Trimethyl borate
 - ≥98%, MW 103.91 g/mol, Sigma-Aldrich
- Processing environments
 - Infiltration and crosslinking: oxygen-free glovebox
 - Pyrolysis: nitrogen atmosphere tube furnace



Ceraset PSZ20



TMB



Isopropanol



Carbon fiber cloth

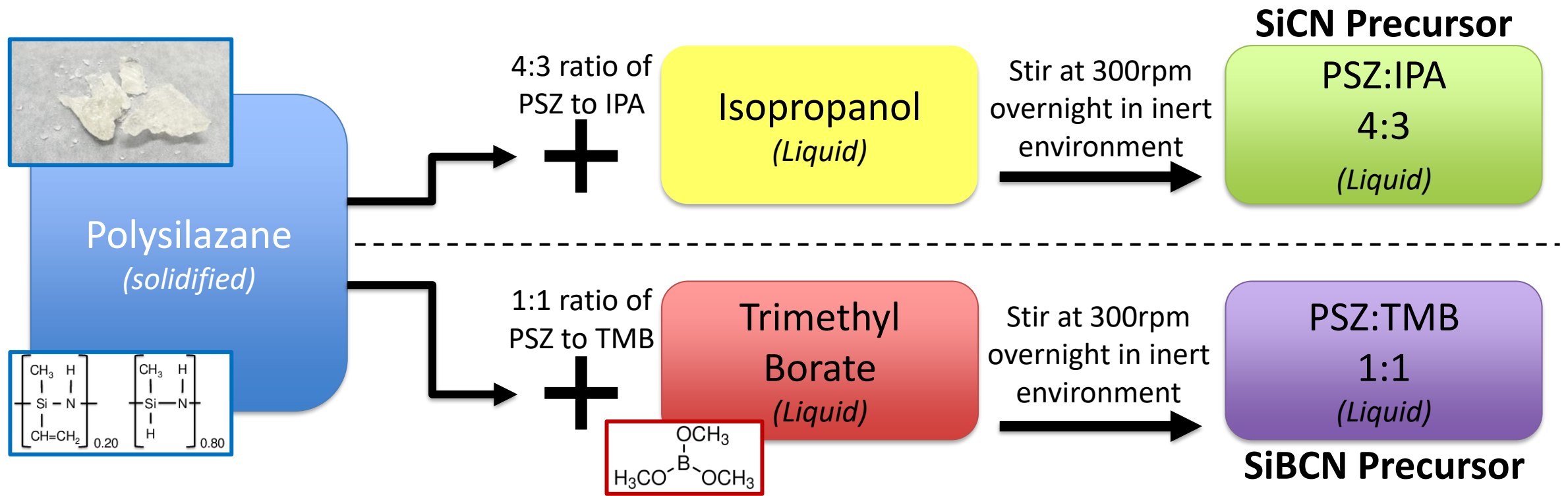


Oxygen-free glovebox



Tube furnace

Liquid Preceramic Polymer Fabrication

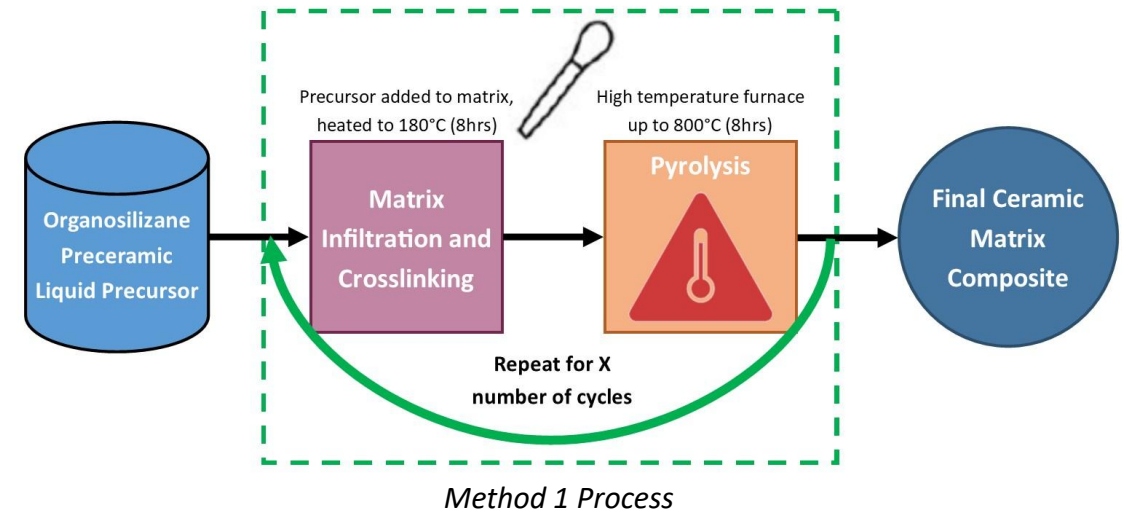


Controlled viscosity → Improved fiber infiltration
Ensured uniform chemical composition prior to conversion

Modified PIP Manufacturing Methods

Base PIP Process:

- Infiltration: 1 μ L:1 mg precursor to reinforcement ratio, added dropwise
- Crosslinking: 180°C for 1 hour, stabilizes polymer
- Pyrolysis: 800°C, 30 minutes in N₂, ceramic conversion

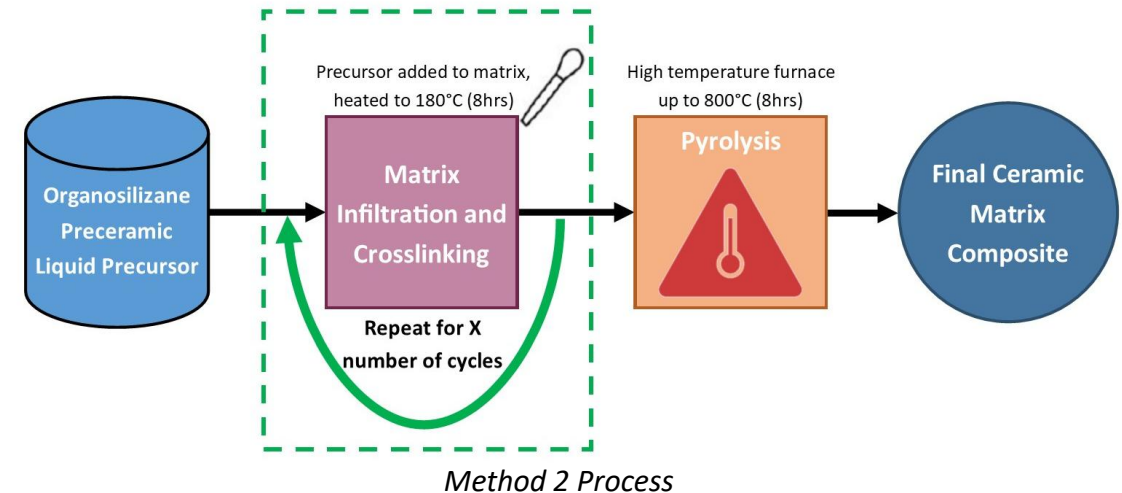


Method 1: Traditional Multi-Cycle PIP

- Repeat full PIP cycle (infiltration, crosslink, pyrolysis)

Method 2: Modified Multi-Cycle PIP

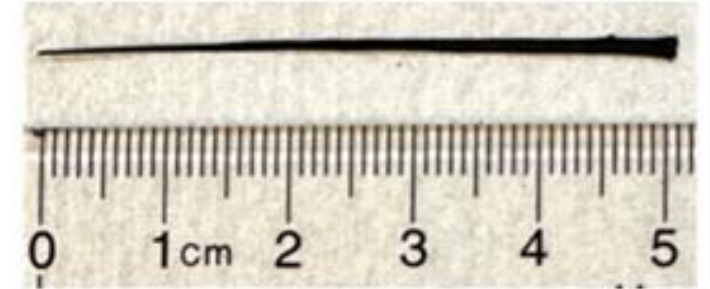
- Only infiltration and crosslinking step is repeated
- Single pyrolysis step for ceramic formation



Experimental Methods

2-phase experiment: Method 1 and Method 2

- Carbon fiber mini-composites: 5 cm bundles
- 10 μ L precursor per infiltration step



METHOD 1

- Multiple cycle conditions
 - Cycle 1, Cycle 2, Cycle 3
- 18 samples
 - 3x SiCN Cycle 1, 3x SiBCN Cycle 1
 - 3x SiCN Cycle 2, 3x SiBCN Cycle 2
 - 3x SiCN Cycle 3, 3x SiBCN Cycle 3
- Mass recorded after each crosslink and pyrolysis

METHOD 2

- Three infiltration/crosslink cycles (pre-pyrolysis), single pyrolysis
- 10 samples
 - 5x SiCN (3 infiltrations, 1 pyrolysis)
 - 5x SiBCN (3 infiltrations, 1 pyrolysis)
- Mass recorded after final crosslink and post-pyrolysis

Cycle-by-Cycle tracking (Method 1) vs. process efficiency evaluation (Method 2)



CHARACTERIZATION AND TESTING

Characterization and Testing Strategy

Purpose

1. Evaluate chemical, structural, and mechanical evolution of CMCs
2. Compare processing methods (Method 1 vs. Method 2)
3. Assess performance for high-temperature applications

Key Outcome:

Relate processing, structure, properties and performance

Characterization Methods

FTIR

Chemical bonding evolution
Polymer to ceramic conversion

XRF

Elemental composition

SEM

Microstructure
Coating quality
Defects

Testing Methods

Mass Analysis

Pyrolysis and ceramic yields
Processing efficiency

Oxidation Testing

High temperature stability

Tensile Testing

Mechanical properties

Fourier Transformation Infrared Spectroscopy

Goal: Track chemical bonding evolution during processing, confirm polymer to ceramic conversion

What is Measured: Vibrational modes of chemical bonds, functional groups (Si-H, N-H, C-H, Si-N, B-N)

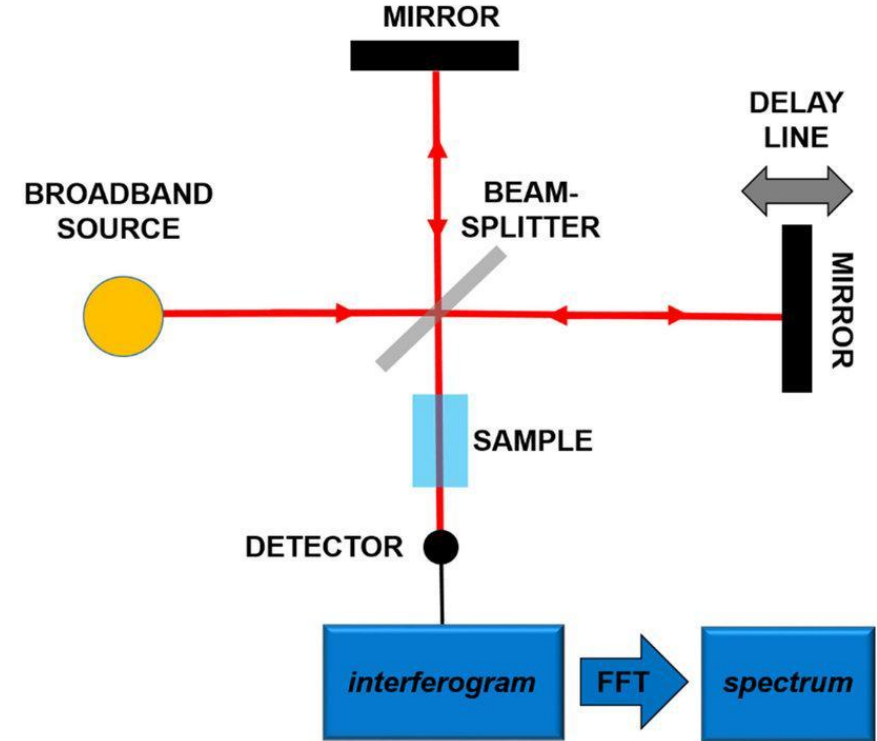
Instrument: Perkin Elmer Spectrum 400 FTIR spectrometer

Method:

- Samples analyzed after precursor synthesis, crosslinking, and pyrolysis
- Spectra collected over standard IR range

Key Outputs:

- Disappearance of organic functional groups (Si-H, C-H, N-H)



X-Ray Fluorescence

Goal: Determine elemental composition

What is Measured: Elemental concentrations (Si, C, N, B, O)

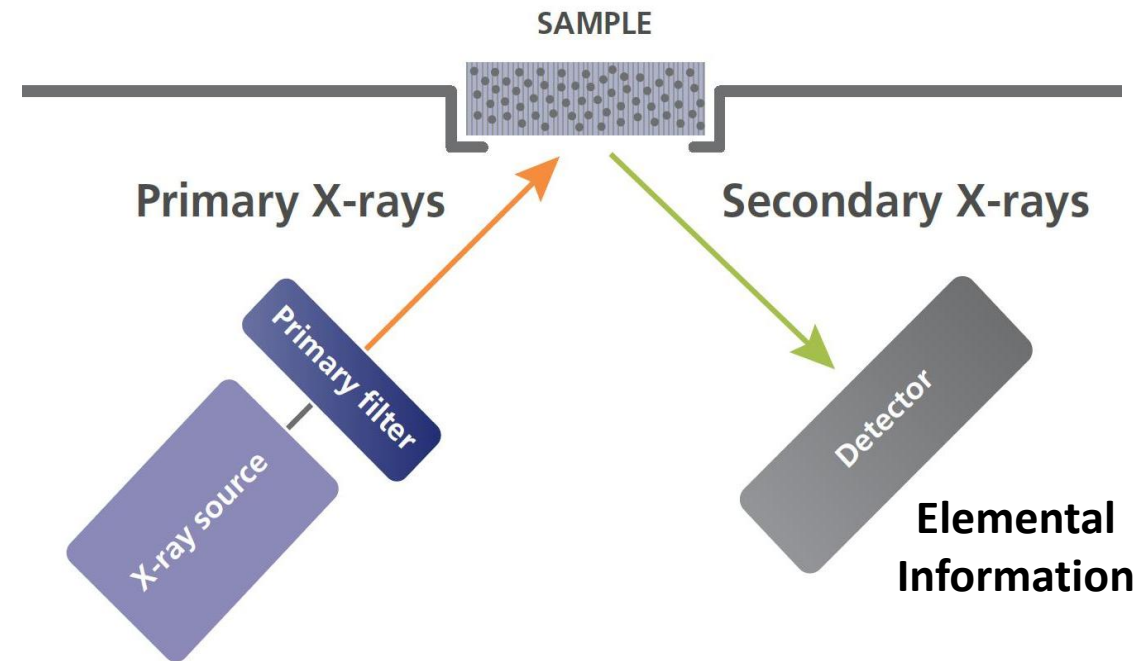
Instrument: Hitachi SU8010, IXRF System

Method:

- Pyrolyzed samples irradiated with X-rays
- Emitted fluorescence measured

Key Outputs:

- Relative elemental composition
- Detection of oxidation



Weight Analysis

Goal: Quantify material conversion efficiency

What is Measured: Mass change through processing steps

$$\text{Ceramic Yield (\%)} = \frac{\text{Total XL Mass Change}}{\text{Total PYRL Mass Change}}$$

$$\text{Carbon Fiber Ratio(\%)} = \frac{\text{Neat CF Mass}}{\text{Final Mass}}$$

$$\text{Ceramic Ratio(\%)} = \frac{\text{Ceramic Mass}}{\text{Final Mass}}$$

Method:

- Weights recorded at:
 - Neat fiber
 - Post-infiltration/crosslink
 - Post-pyrolysis
- Pyrolysis yield and ceramic yield calculated

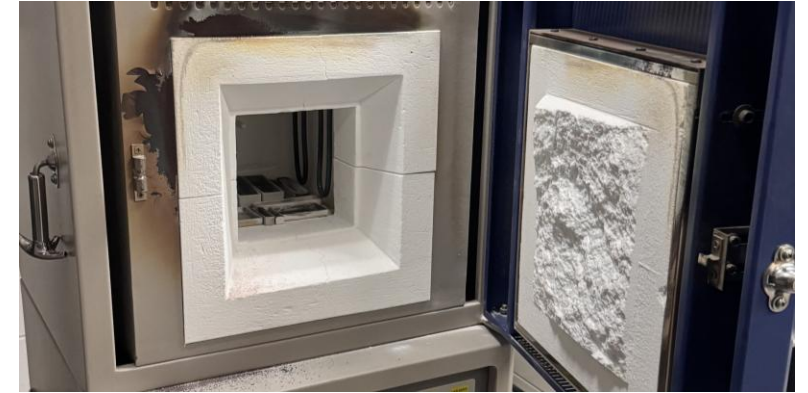
Key Outputs:

- Pyrolysis yield
- Ceramic yield
- Comparison of method 1 and method 2 processing
- Insight to matrix formation and densification

Oxidation Testing

Goal: Evaluate high-temperature oxidation resistance

What is Measured: Mass retention after oxidation exposure



Instrument: SentroTech box furnace

Method:

- Pyrolyzed samples exposed to elevated temperature (800C) in air
- Mass recorded before and after

Key Outputs:

- Weight retention (%)
- Comparison to uncoated carbon fiber
- Effect on cycle number and manufacturing method
- Oxidation diffusion resistance

Scanning Electron Microscopy (SEM)

Goal: Analyze microstructure and coating quality

What is Measured: Surface morphology and microstructure features

Instrument: Carl Zeiss EVO MA10

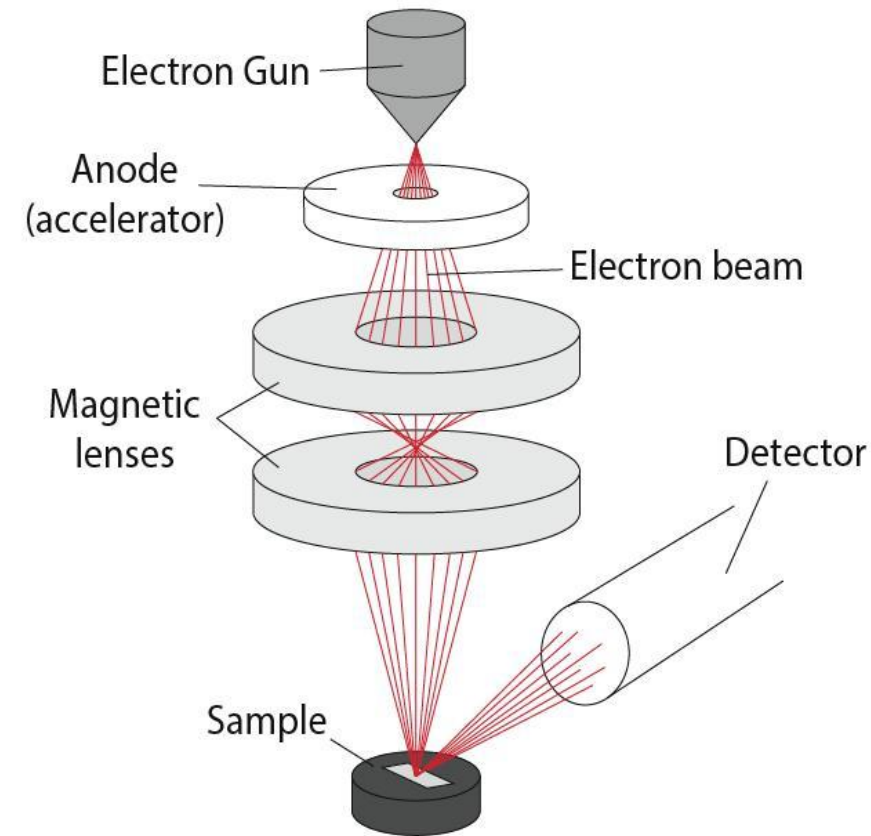
Method:

- Imaging pyrolyzed and post-oxidation samples
- High-magnification electron imaging

Key Outputs:

- Fiber coating coverage
- Matrix uniformity
- Cracks and porosity
- Oxidation damage mechanisms

Scanning Electron Microscope (SEM)



Mechanical Testing

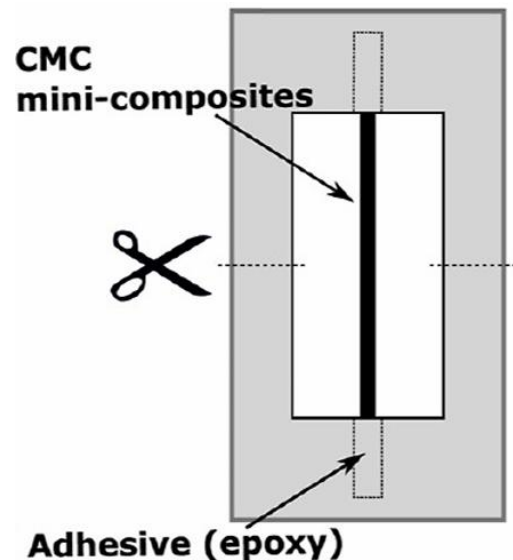
Goal: Evaluate mechanical performance

What is Measured: Stress-Strain response

Instrument: Shimadzu Universal Testing machine

Method:

- Cardstock mounts with thin layer of adhesive for bonding and painters tape to improve grip and reduce slippage during testing
- Uniaxial tensile loading at 0.2mm/minute
- Invalid samples removed (slippage/grip failure)
- Mechanical values calculated



$$\sigma = F / A$$

Engineering stress

$$\varepsilon = \Delta L / L_0$$

Engineering strain

$$U_T = \int_0^{\varepsilon_f} \sigma d\varepsilon$$

Ultimate tensile strength

Key Outputs:

- Elastic modulus
- Ultimate tensile strength
- Toughness
- Failure strain
- Comparison: neat CF vs. method 2 SiCN vs. SiBCN

RESULTS: METHOD 1

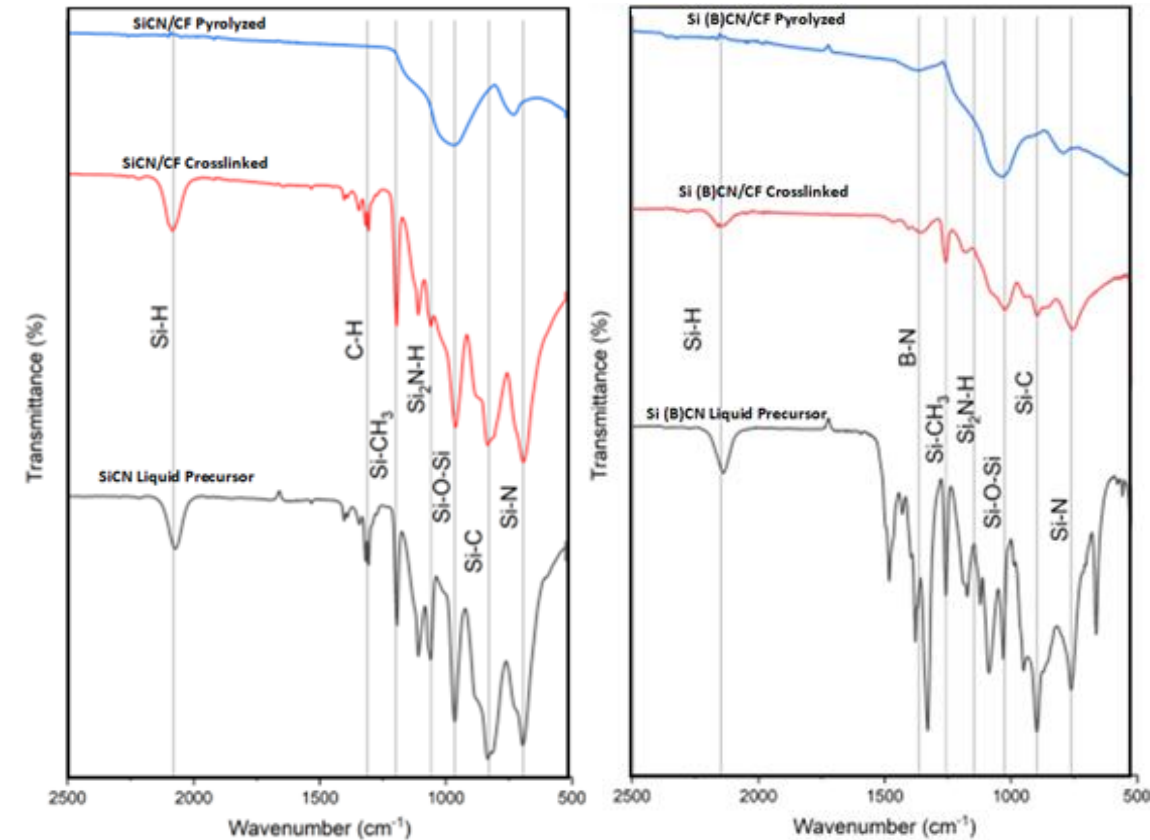
FTIR Results: Method 1

Observations

- Precursor peaks
 - Si-CH₃
 - Si-H
- Crosslinked:
 - Reduction in Si-H
- Pyrolyzed
 - Strong Si-O-Si
- Si-C
- Reduced organic groups (C-H, Si-H)
- SiBCN shows strong B-N bonding peak development
- Minimal differences between cycles

Conclusions:

- Confirms successful polymer to ceramic conversion
- Organic decomposition
- Cycle number does not significantly change chemistry



Additional PIP cycles affect structure, not composition

Mass Analysis Results: Method 1

All masses in milligrams

Type	Total Number of Cycle	Neat CF Mass	1 st Cycle				2 nd Cycle				3 rd Cycle			
			XL Mass	PYRL Mass	XL Mass Change	PYRL Mass Change	XL Mass	PYRL Mass	XL Mass Change	PYRL Mass Change	XL Mass	PYRL Mass	XL Mass Change	PYRL Mass Change
SiCN	1	10.3	18.7	16.1	8.4	2.6								
	2	9.8	19.5	16.4	9.7	3.1	23.4	21.2	7	2.2				
	3	9.8	24.3	19.8	14.5	4.5	29.3	25.8	9.5	3.5	32.1	29.9	6.3	2.2
SiBCN	1	10.2	12.8	12.4	2.6	0.4								
	2	9.8	12.6	12.2	2.8	0.4	15.3	14.3	3.1	1				
	3	9.8	13.2	12.6	3.4	0.6	15.9	15	3.3	0.9	17	16.3	2	0.7

Observations:

- As number of cycles increases, preceramic material added during infiltration and crosslinking decreases
 - Indicates progressive pore filling and reduced infiltration volume
- Less added preceramic material associates to a smaller pyrolysis mass change
- SiCN shows significantly higher polymer loading than SiBCN

Mass Analysis Results: Method 1

All masses in milligrams

Type	Total Number of Cycles	1 st Cycle Ceramic Yield (%)	2 nd Cycle Ceramic Yield (%)	3 rd Cycle Ceramic Yield (%)	Final Mass	Total Mass Change	Final Ceramic Yield (%)	Carbon Fiber Ratio (%)	Ceramic Ratio (%)
SiCN	1x	69.05%			16.1	5.8	69.05%	63.98%	36.02%
	2x	68.04%	68.57%		21.2	11.4	68.26%	46.23%	53.77%
	3x	68.97%	63.16%	65.08%	29.9	20.1	66.34%	32.78%	67.22%
SiBCN	1x	84.62%			12.4	2.2	84.62%	82.26%	17.74%
	2x	85.71%	32.26%		14.3	4.5	76.27%	68.53%	31.47%
	3x	82.35%	27.27%	65.00%	16.3	6.5	74.71%	60.12%	39.88%

Observations:

- SiCN adds significantly more total ceramic mass compared to SiBCN for all cycles
- SiCN ceramic yield is more consistent (63-69%), SiBCN ceramic yield shows more variability
- SiCN is **ceramic** dominated, SiBCN is **fiber** dominated

$$\text{Ceramic Yield (\%)} = \frac{\text{Total XL Mass Change}}{\text{Total PYRL Mass Change}}$$

$$\text{Carbon Fiber Ratio (\%)} = \frac{\text{Neat CF Mass}}{\text{Final Mass}}$$

$$\text{Ceramic Ratio (\%)} = \frac{\text{Ceramic Mass}}{\text{Final Mass}}$$

SEM Analysis Results: Method 1

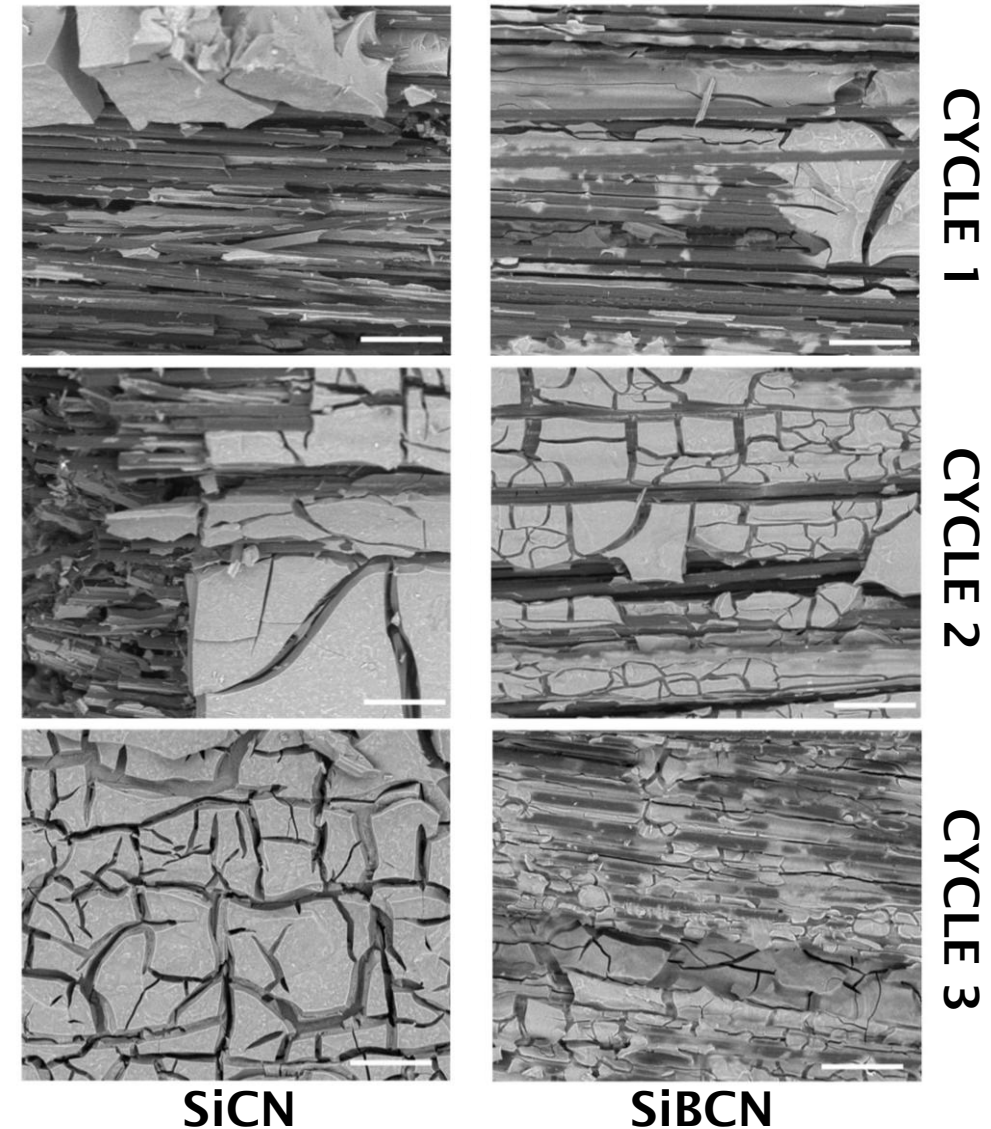
Observations

- Clear ceramic coating formation on fibers
- Low cycle samples: thin coatings, exposed fibers
- High cycle samples: thicker, more continuous coatings
- Non-uniform coating thickness observed
- Evidence of microcracking and surface irregularities

Conclusions:

- Increasing cycles improved fiber coverage and matrix formation
- Repeated pyrolysis introduces shrinkage stresses which leads to matrix cracking

**Quality improves, but defects
accumulate with cycles**

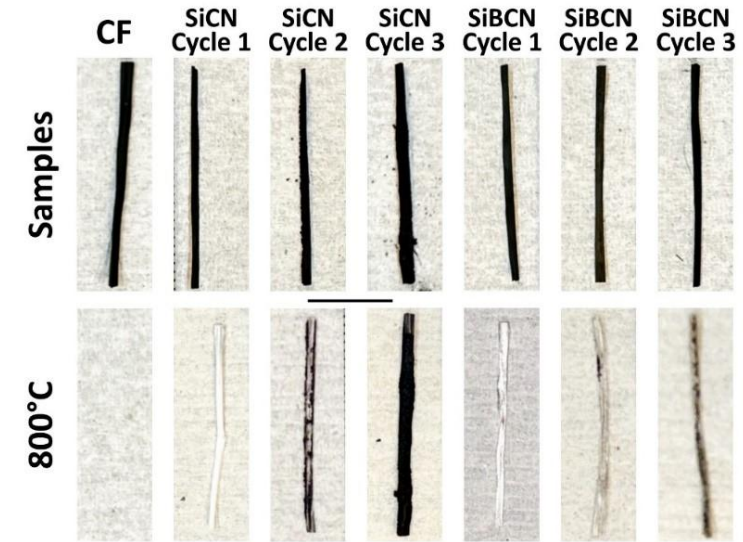


1500x magnification, Scale bar is 50 μ m

Oxidation Testing Results: Method 1

All masses in milligrams

Type	Number of Cycles	Neat CF Mass	Final Mass (Post PYRL)	Ceramic Mass	Post Oxidation Mass	Mass Loss from Oxidation	Mass Retention (%)	Mass Retention Efficiency (%)
SiCN	1x	10.3	16.1	5.8	4.4	11.7	27.33%	75.86%
	2x	9.8	21.2	11.4	10.2	11.0	48.11%	89.47%
	3x	9.8	29.9	20.1	20.8	9.1	69.57%	103.48%
SiBCN	1x	10.2	12.4	2.2	2.4	10.0	19.35%	109.09%
	2x	9.8	14.3	4.5	4.2	10.1	29.37%	93.33%
	3x	9.8	16.3	6.5	5.9	10.4	36.20%	90.77%



Observations:

- Mass retention increases with cycle number for both materials
- SiCN exhibits higher overall oxidation resistance
- SiBCN shows more consistent weight retention efficiency
- Efficiency trends diverge with cycle number

$$\text{Mass Retention}(\%) = \frac{\text{Post Oxidation Mass}}{\text{Final Mass}}$$

$$\text{Mass Retention Efficiency}(\%) = \frac{\text{Post Oxidation Mass}}{\text{Ceramic Max}}$$

Conclusions:

- Ceramic loading dominates oxidation performance
- SiBCN provides more efficient protection at low ceramic loading

RESULTS: METHOD 2

Mass Analysis Results: Method 2

All masses in milligrams

Sample	Neat Carbon Fiber (CF) Mass	Crosslinking (XL)			Pyrolysis (PYRL)		Total Mass Change
		3 rd XL Mass	Total XL Mass Change	Pre-PYRL Polymer Ratio (%)	Post-PYRL Mass	Post-PYRL Mass Change	
SiCN_1	10.2	31.9	21.7	212.75%	25.7	6.2	15.5
SiCN_2	9.9	30.4	20.5	207.07%	25.5	4.9	15.6
SiCN_3	9.7	25.8	16.1	165.98%	22.3	3.5	12.6
SiCN_4	9.6	31.4	21.8	227.08%	24.3	7.1	14.7
SiCN_5	10.0	28.7	18.7	187.00%	23.2	5.5	13.2
SiBCN_1	9.7	17.0	7.3	75.26%	12.3	4.7	2.6
SiBCN_2	10.0	18.2	8.2	82.00%	13.2	5.0	3.2
SiBCN_3	9.8	18.0	8.2	83.67%	12.5	5.5	2.7
SiBCN_4	9.8	18.3	8.5	86.73%	13.5	4.8	3.7
SiBCN_5	10.1	17.6	7.5	74.26%	14.3	3.3	4.2

Total XL Mass Change = 3rd XL Mass – Neat CF Mass

PostPYRL Mass Change = PostPYRL Mass – 3rd XL Mass

PrePYRL Polymer Ratio = $\frac{\text{Total XL Mass Change}}{\text{Neat CF Mass}}$

Total Mass Change = PostPYRL Mass – Neat CF Mass

Mass Analysis Results: Method 2

$$\text{Ceramic Yield (\%)} = \frac{\text{Total XL Mass Change}}{\text{Total PYRL Mass Change}}$$

$$\text{Carbon Fiber Ratio(\%)} = \frac{\text{Neat CF Mass}}{\text{Final Mass}}$$

$$\text{Ceramic Ratio(\%)} = \frac{\text{Ceramic Mass}}{\text{Final Mass}}$$

Observations

- SiCN shows significantly higher polymer uptake
- Final pyrolyzed mass is almost 4x higher for SiCN
- Ceramic fraction is much higher for SiCN

Conclusions

- SiCN achieves much higher ceramic loading
- SiBCN shows limited ceramic incorporation
- SiBCN is fiber dominated, which will likely be reflected in oxidation and mechanical testing results

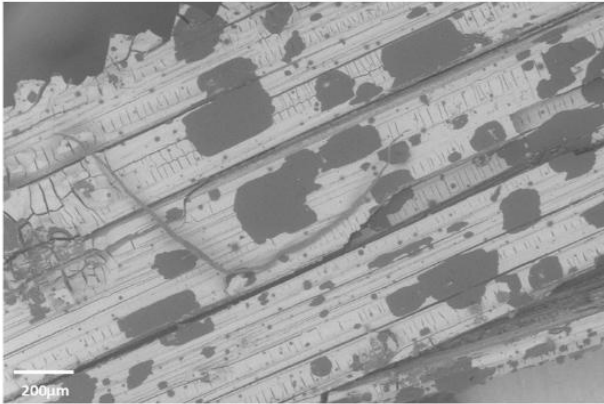
Sample	Final Mass (Post PYRL)	Ceramic Mass	Ceramic Yield (%)	Carbon Fiber Ratio (%)	Ceramic Ratio (%)
SiCN_1	25.7	15.5	71.43%	39.69%	60.31%
SiCN_2	25.5	15.6	76.10%	38.82%	61.18%
SiCN_3	22.3	12.6	78.26%	43.50%	56.50%
SiCN_4	24.3	14.7	67.43%	39.51%	60.49%
SiCN_5	23.2	13.2	70.59%	43.10%	56.90%
SiBCN_1	12.3	2.6	35.62%	78.86%	21.14%
SiBCN_2	13.2	3.2	39.02%	75.76%	24.24%
SiBCN_3	12.5	2.7	32.93%	78.40%	21.60%
SiBCN_4	13.5	3.7	43.53%	72.59%	27.41%
SiBCN_5	14.3	4.2	56.00%	70.63%	29.37%

SiCN: Ceramic-dominated ($\approx 60\%$)

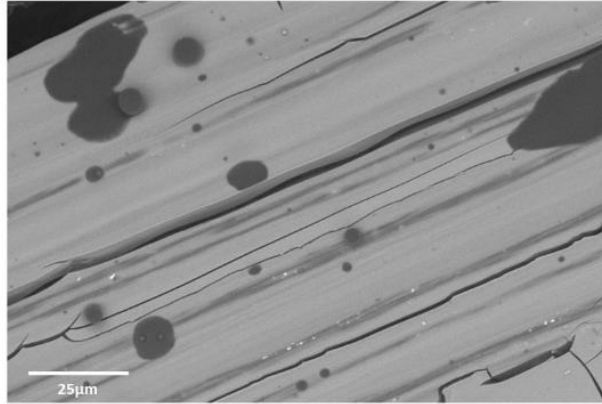
SiBCN: Fiber-dominated ($\approx 75\%$)

SEM Analysis Results: Method 2 (Pyrolyzed)

SiCN-CF (pyrolyzed)

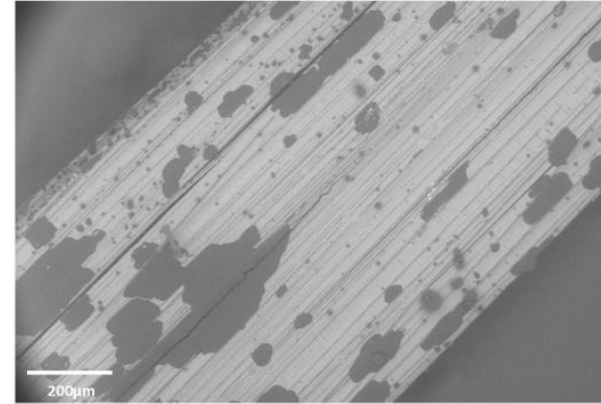


a.) Center, 260x

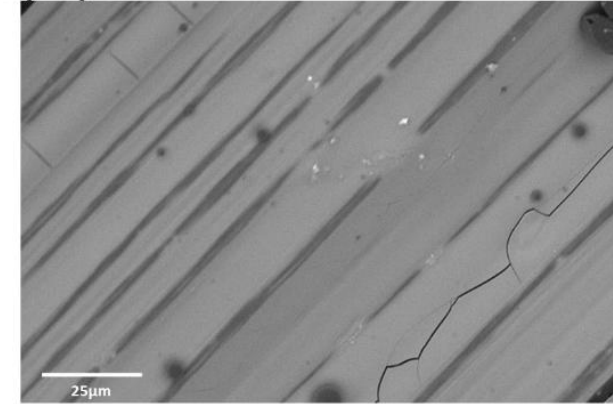


b.) Center, 2950x

SiBCN-CF (pyrolyzed)



a.) Center, 250x



b.) Center, 2950x

Observations

- Clear ceramic coating formation on fibers
- Coating appears uniform and continuous
- Dark circle regions: polymer-rich pooling before pyrolysis, resulting in ceramic concentration after pyrolysis
- Microcrack regions shown, attributed to shrinkage during polymer-to-ceramic conversion

Conclusions:

- SiCN:
 - Slightly more visible cracking
 - More pronounced ceramic pooling
- SiBCN
 - More uniform fiber coverage
 - Cleaner fiber definition, suggests improved wetting or distribution

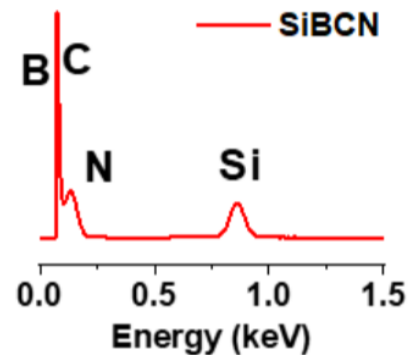
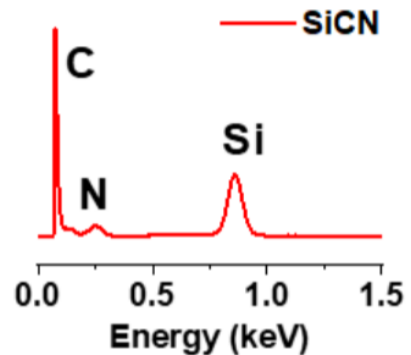
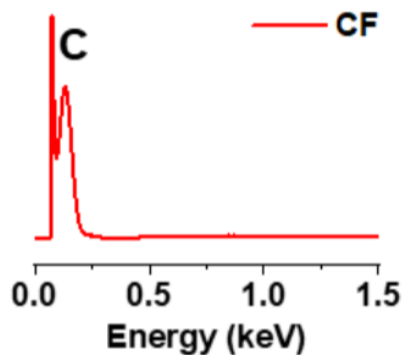
XRF Results: Method 2

Observations

- Neat CF dominated by strong carbon peak
- Both coated samples show clear evidence of additional ceramic-related elements, in addition to carbon peaks related to the carbon fiber reinforcement
- SiBCN shows strong boron signal

Conclusions:

- Ceramic-coated composite system is produced
- Appearance of Si and N in both samples confirm formation of SiCN-based ceramic phases
- Boron signal in SiBCN confirms incorporation into final material



Confirms Method 2 successfully converts samples into intended ceramic chemistries

Oxidation Testing Results: Method 2

All masses in milligrams

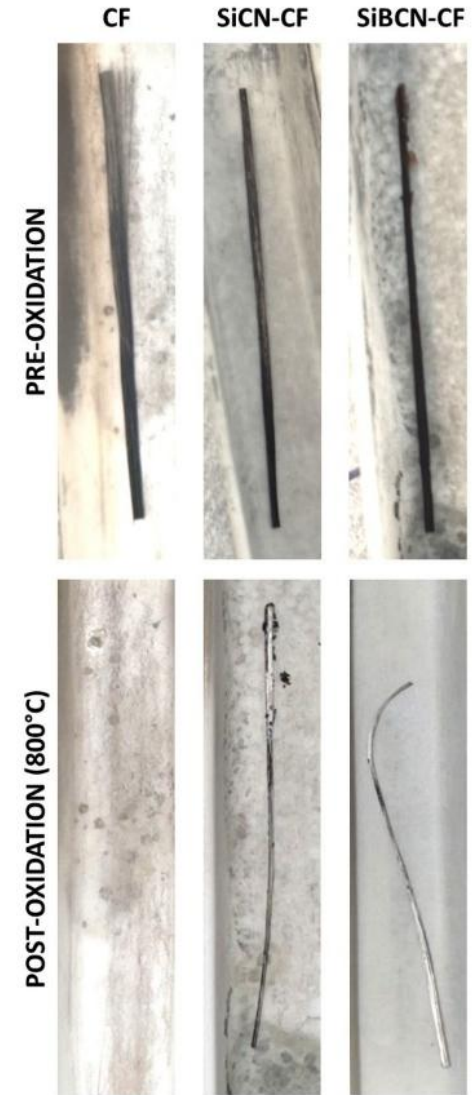
Type	Neat CF Mass	Final Mass (Post PYRL)	Ceramic Mass	Post Oxidation Mass	Mass Loss from Oxidation	Mass Retention (%)	Mass Retention Efficiency (%)
SiCN	10	23.2	13.2	8.4	4.8	36.21%	63.64%
SiBCN	10.1	14.3	4.2	3.2	1	22.38%	76.19%

Observations

- SiCN exhibits higher overall oxidation resistance
- SiBCN shows higher mass retention efficiency
- Samples exhibited visible curling/warping

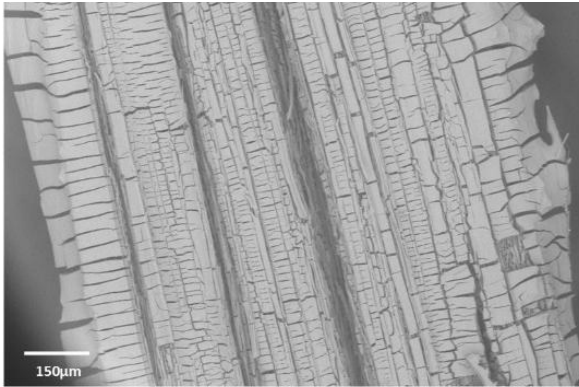
Conclusions:

- Absolute oxidation performance is strongly influenced by ceramic loading
- SiBCN has much more efficient oxidation protection per unit ceramic
- Sample curling indicates presence of residual thermal stresses, likely caused by fiber-matrix expansion mismatches

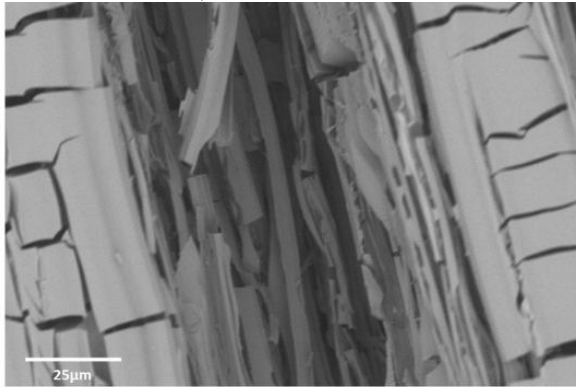


SEM Analysis Results: Method 2 (Oxidized)

SiCN-CF (oxidized)

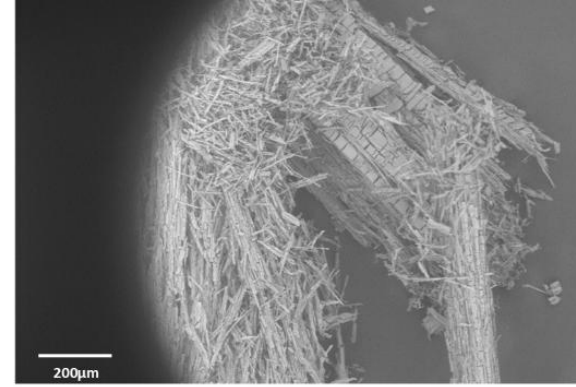


a.) Center, 230x

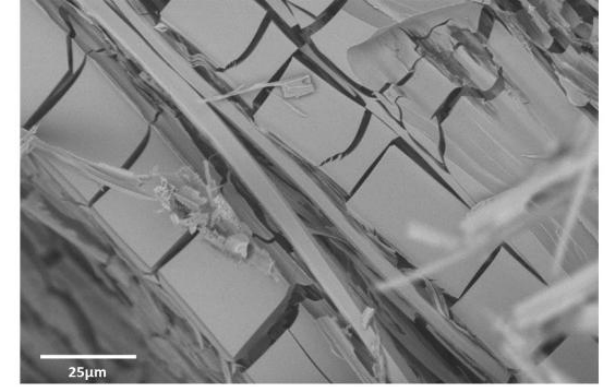


b.) Center, 2950x

SiBCN-CF (oxidized)



a.) Center, 225x



b.) Center, 2950x

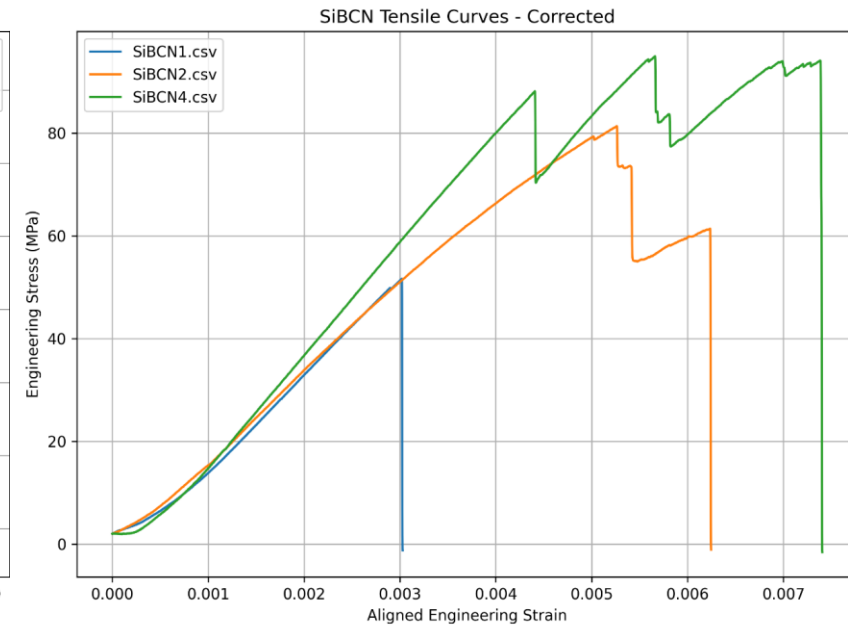
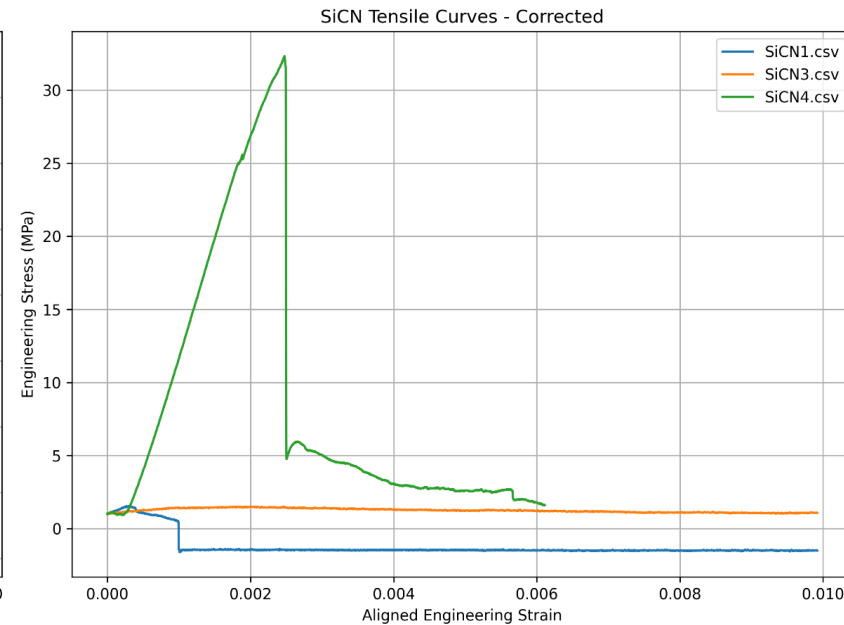
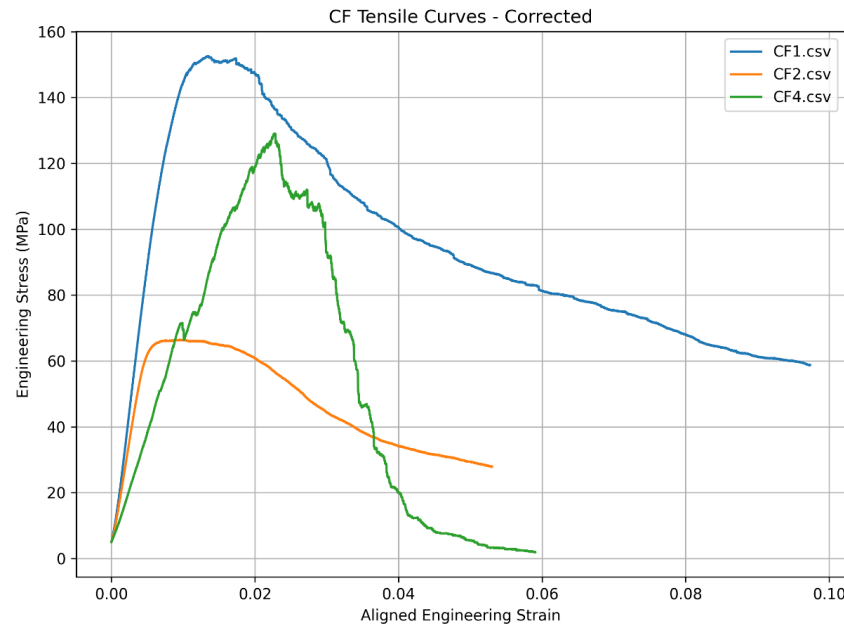
Observations

- Surfaces show continuous, cracked oxide scale, likely formation of SiO_2 protective layer
- Cracking pattern is segmented, typical of thermal expansion mismatch during oxidation
- Fibers remain present and structurally intact, indicated coating continues to shield fiber surfaces
- Cracking is more pronounced than pyrolyzed state
- Cracks do not appear to fully penetrate fibers

Conclusions:

- SiCN:
 - More continuous oxide scale
 - Better overall surface protection
- SiBCN
 - More fiber exposure and fragmentation
 - Suggests slightly reduced oxidation resistance

Tensile Testing Results: Method 2



Observations

- Neat CF has highest UTS and largest failure strain
- SiCN samples 1 and 3 showed premature failure, possibly due to grip and slippage effects
- SiBCN samples show step-like drops in curves, likely due to progressive failure and individual fiber breakage

Conclusions:

- Addition of ceramic matrix results in increased stiffness and decreased ultimate strength and strain, as expected
- SiBCN shows better load transfer and matrix interaction compared to SiCN

Tensile Testing Results: Method 2 vs. Single Cycle

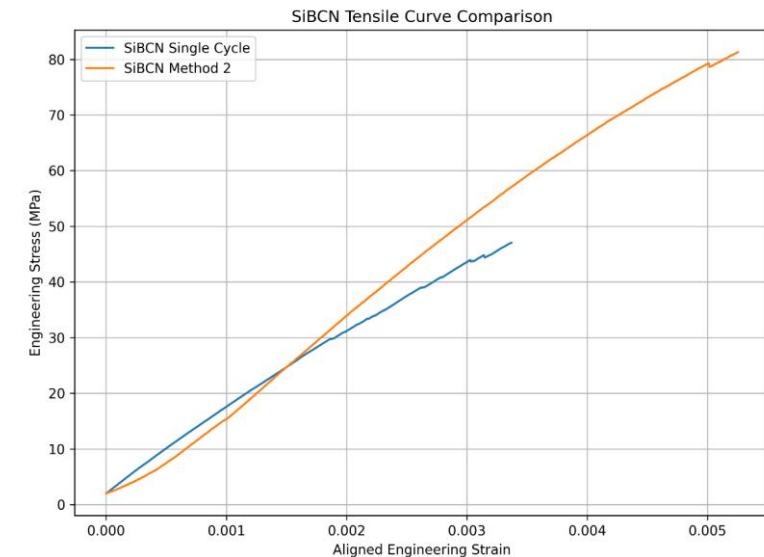
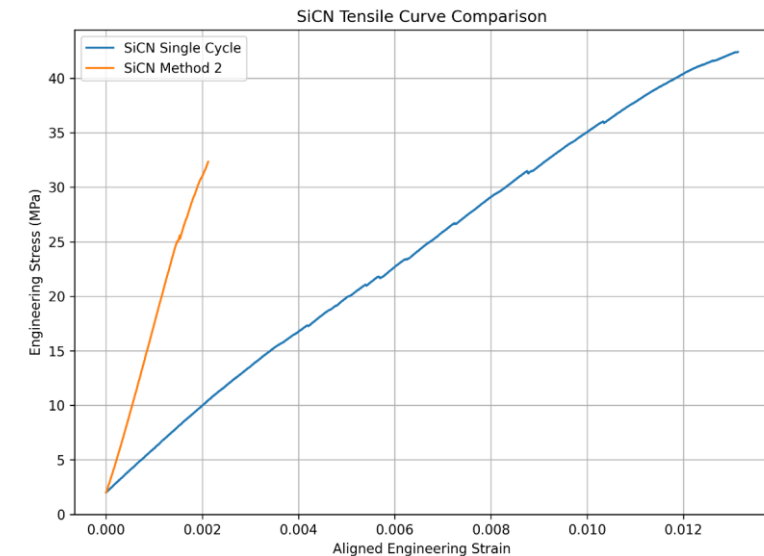
Material	Method	Elastic Modulus (GPa)	UTS (MPa)	Toughness (MJ/m ³)
SiCN	Single Cycle	3.263	42.415	0.317
	Method 2	16.100	32.335	0.038
SiBCN	Single Cycle	14.833	47.046	0.089
	Method 2	17.983	81.313	0.228

Observations

- SiCN
 - Elastic modulus increases significantly, UTS and toughness decreases
 - Stress-strain response is steeper with early failure
- SiBCN
 - Elastic modulus, UTS, and toughness increase
 - Stress-strain response shows improved load capacity

Conclusions:

- SiCN
 - Additional infiltration improves stiffness
- SiBCN
 - Additional infiltration improves matrix uniformity and load transfer
 - Higher strength and toughness



CONCLUSIONS

Method Comparison: Mass and Oxidation Analysis

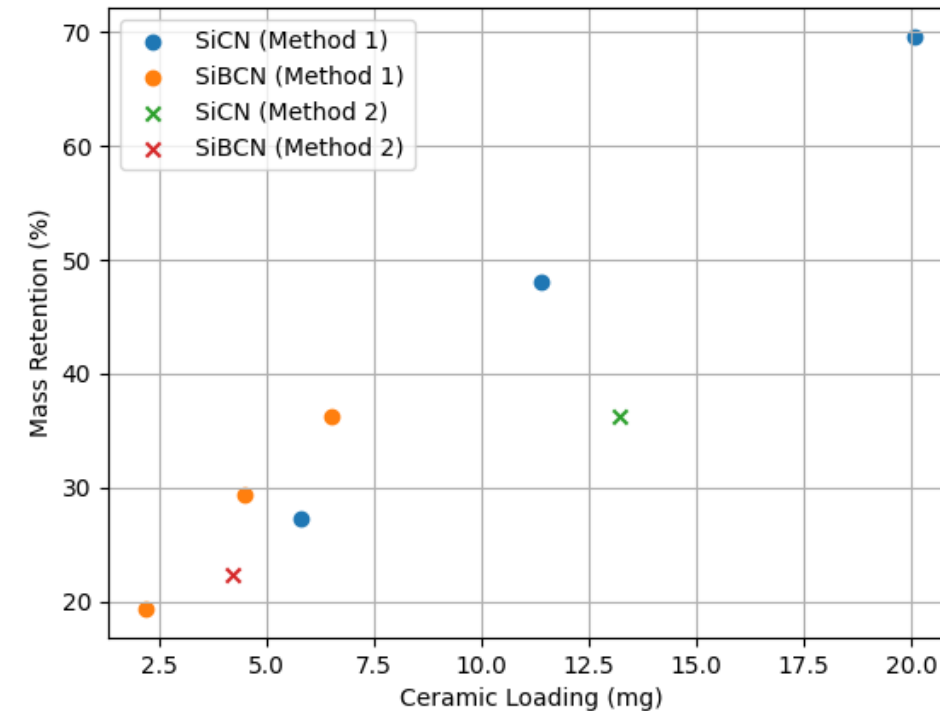
Observations

- Ceramic loading directly correlates with oxidation performance
- SiCN consistently achieves higher ceramic loading resulting in improved absolute oxidation resistance
- SiBCN achieves comparable retention at lower ceramic loading and more efficient oxidation resistance per unit ceramic
- Method 1 introduces cycle-dependent behavior
- Method 2 isolates intrinsic material behavior, providing clearer efficiency comparison

Conclusions

- Oxidation performance is dominated by ceramic loading
- SiBCN provides more efficient oxidation protection per unit ceramic (especially evident at low ceramic loading)
- Processing method strongly influences observed performance
 - Method 1 → cumulative buildup and densification effects
 - Method 2 → direct comparison of material efficiency

Ceramic Loading vs. Oxidation Mass Retention



Method Comparison: Processing Time

Processing Efficiency

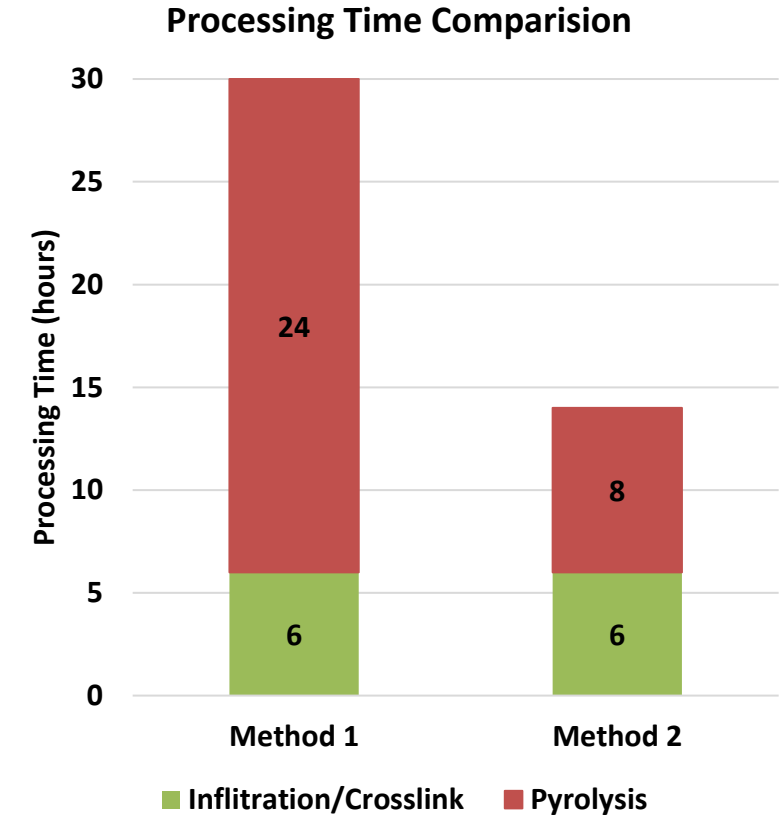
- Method 2 reduces processing time: 30 hrs → 14 hrs (~53% reduction)
- Time savings driven by eliminating repeated pyrolysis

Key Insight

- Additional pyrolysis cycles in Method 1 provide limited performance gains
- Method 2 achieves comparable performance when accounting for reduced ceramic loading

Implications for Manufacturing

- Reduced processing time → increased scalability
- Lower cost and higher throughput
- More viable for aerospace applications



Modified PIP (Method 2) enables a scalable, time-efficient manufacturing route for CMCs without sacrificing critical performance

Performance vs. Processing Tradeoff

Method 1 (Traditional Multi-Cycle PIP)

- High ceramic loading → highest absolute oxidation resistance
- Progressive densification with cycles
- Diminishing returns with additional cycles
- Increased processing time and defect accumulation risk

Method 2 (Modified Single Pyrolysis PIP)

- Lower ceramic loading, but higher efficiency per unit ceramic
- Uniform coating achieved prior to pyrolysis
- Comparable performance relative to reduced loading
- Significantly reduced processing time (~53%)

Performance is governed by both ceramic loading and efficiency: Method 1 maximizes loading, while Method 2 maximizes efficiency and scalability.

Final Conclusions and Engineering Impact

What This Work Demonstrated

- Development of a single-pyrolysis modified PIP process
- Direct comparison to traditional multi-cycle PIP
- Evaluated across:
 - Ceramic yield
 - Oxidation resistance
 - Microstructure
 - Mechanical behavior

Key Technical Findings

- Ceramic loading drives absolute oxidation performance
- SiBCN provides higher efficiency per unit ceramic
- Coating continuity is more important than repeated pyrolysis
- Multi-cycle processing shows diminishing returns

Engineering Impact

- Modified PIP reduces processing time by ~53%
- Eliminates repeated high-temperature processing bottleneck
- Enables scalable, cost-effective CMC manufacturing

Modified PIP (Method 2) provides a scalable manufacturing route that maintains strong oxidation performance by optimizing efficiency rather than maximizing ceramic loading.

Future Work

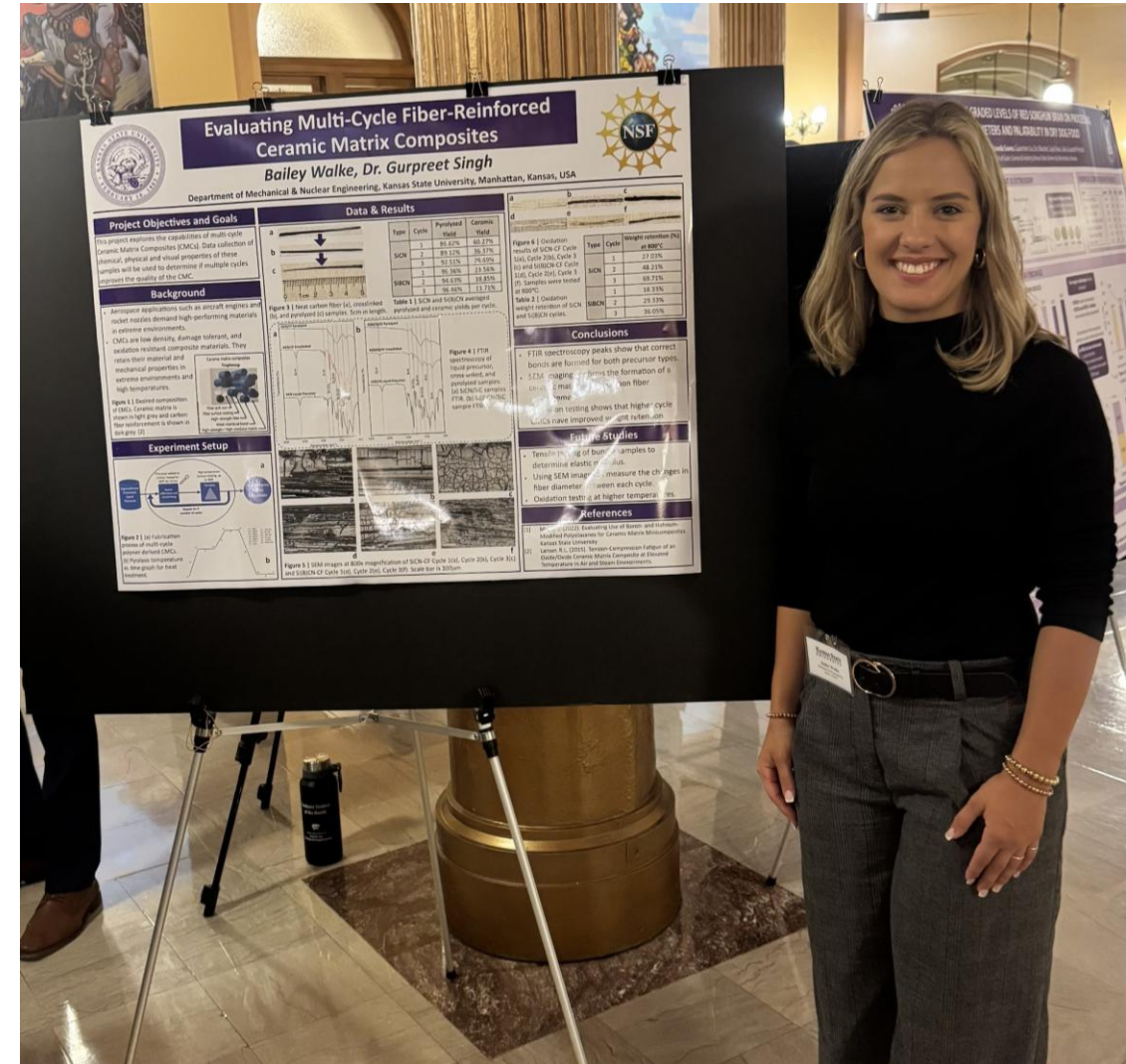
- Quintenary systems (SiHfBCN)
- Ceramic reinforcement materials (SiC)
- Complex geometries
- Further mechanical testing (4 point bend, tensile)
- Cycle number optimization study
- Advanced infiltration techniques
- Pyrolysis optimization study

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THANK YOU!

QUESTIONS?



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